CS532 Quiz 6: Structure from Motion 2

Name: _____

Question 1 of 2. Given about 500,000 photos of Manhattan with weakly overlapping capture, pick either an incremental, global, hierarchical, or a hybrid approach to reconstruct the city and defend your choice.

Solution:

- For a large dataset such as this, a pure incremental pipeline would suffer from the $O(N^3)$ complexity of bundle adjustment.
- Manhattan has lots of repetitive structures such as similar looking windows and facades, making global SfM vulnerable to bad pairwise geometry.
- A hierarchical approach lets you restrict global or incremental optimization to smaller better conditioned regions, and then merge them.

Question 2 of 2. You later find out that these images given to you are crowd-sourced (i.e. contaminated with WTFs). Suggest a simple change to your RANSAC estimator in the geometric verification stage of your chosen SfM pipeline to make it more robust to contaminated images.

Solution: One simple mpodification would be as follows:

- After RANSAC fits your two-view model, analyze the inlier set. Check if a large fraction of inliers lie in a thin band near the image borders (where watermarks/frames usually are), and whether those inliers are explained well by a pure 2D translation (i.e., all inlier displacements are nearly identical).
- If this "border-translation" pattern holds, reject the RANSAC model and the image pair as a WTF pair, even if the inlier count is high